



M3200

Pressure Transducer

SPECIFICATIONS

- Analog Outputs (V/mA)
- 14-Bit Digital Output for Pressure and 11-Bit for Temperature
- CE Compliance
- Weatherproof
- 0.5% zero offset, 1.5 % accuracy (Total error band)

The M3200 pressure transducer from the Microfused line of TE is suitable for measurement of liquid or gas pressure, even for difficult media such as contaminated water, steam, and mildly corrosive fluids.

The transducer pressure cavity is machined from a solid piece of 17-4PH stainless steel. The standard version includes a 1/4 NPT pipe thread allowing a leak-proof, all metal sealed system. With excellent durability, there are no welds or organics exposed to the pressure media.

TE's proprietary Microfused technology, derived from demanding aerospace applications, employs micromachined silicon piezoresistive strain gages fused with high temperature glass to a stainless-steel diaphragm. This approach achieves media compatibility simply and elegantly while providing an exceptionally stable sensor without the PN junctions of conventional micromachined sensors.

This product is geared towards industrial and commercial OEMs for small to high volume applications. Standard configurations are suitable for many applications. Please contact factory for your customization needs.

FEATURES

- One Piece Stainless Steel Construction
- Digital Pressure and Temperature Output or Analog mV/Amplified Output
- Compact
- 17-4PH Stainless Steel
- Customizable

APPLICATIONS

- Pumps and Compressors
- Hydraulic/Pneumatic Systems
- Automotive Test Systems
- Energy and Water Management
- Medical Gas Pressure
- Leak Detection
- Remote Measuring Systems
- General Pressure Measurements

STANDARD RANGES

Range (psi)	Range (bar)	Gage/Compound
0 to 100	0 to 007	•
0 to 250	0 to 017	•
0 to 500	0 to 035	•
0 to 01k	0 to 070	•
0 to 2k5	0 to 170	•
0 to 05k	0 to 350	•
0 to 7k5	0 to 500	•
0 to 10k	0 to 700	•

PERFORMANCE SPECIFICATIONS (ANALOG)

Unless otherwise specified: All parameters measured at 25°C

PARAMETERS	MIN	TYP	MAX	UNITS	NOTES
Accuracy (Combined linearity, hysteresis & repeatability)	-0.25		0.25	% F.S BFSL	
Zero offset	-0.5		0.5	%F.S.	@ 25°C
Pressure Cycles	1.0E+6			0~F.S. Cycles	
Proof Pressure	2X			Rated	
Burst Pressure	5X			Rated	≤20kpsi
Isolation, Body to Any Lead	50			MΩ	@ 250V _{DC}
Load Resistance (R _L)	>100			kΩ	Voltage Output
Load Resistance	<(Supply Voltage-9V)/0.02A			Ω	Current Output
Current Consumption			5	mA	Voltage Output
Dielectric Strength			2	mA	@500 V _{AC} 1 min
Long Term Stability (1 year)	-0.25		0.25	%Span	
Total Error Band	-1.5		1.5	%F.S.	Over comp. temp
Compensated Temperature	-20		85	°C	
Operating Temperature	-40		125	°C	Except Cable 105°C max
Storage Temperature	-40		125	°C	Except Cable 105°C max
Weather proof Rating	IP67 for cable & M12 type, IP66 for Packard type, IP65 for Form C type				Note 1
Rise Time (10% - 90%)	<2 ms (mV Output); <3ms (mA Output)				
Wetted Material	17-4PH Stainless Steel				
Shock	50g, 11 msec Half Sine Shock per MIL-STD-202G, Method 213B, Condition A				
Vibration	±20g, MIL-STD-810C, Procedure 514.2-2, Curve L				

Compliances⁶

EN 55022 Emissions Class A & B

IEC 61000-4-2 Electrostatic discharge immunity (4kv contact / 8kv air discharge)

IEC 61000-4-3 Radiated, Radio-Frequency Electromagnetic field immunity (10 V/m; 80M-1GHz; 3 V/m, 1.4 – 2.0GHz; 1 V/m, 2.0 – 2.7GHz)

IEC 61000-4-4 Electrical Fast Transient/Burst Immunity (±1kV)

IEC 61000-4-5 Surge (line to line: ±1.0kV/42Ω; Line to case: ±1.0kV/42Ω)

IEC 61000-4-6 Immunity to conducted disturbances, induced by radio-frequency fields (150k-80MHz, 3V_{RMS} for current output model, 10V_{RMS} for voltage model)

PERFORMANCE SPECIFICATIONS (DIGITAL)

Unless otherwise specified: All parameters measured at 25°C & 3.3VDC

PARAMETERS	MIN	TYP	MAX	UNITS	NOTES
Output at Zero Pressure	750	1000	1250	Count	
Output at FS Pressure	14720	15000	15250	Count	
Current Consumption			3.5	mA	
Current Consumption (sleep mode)			5	µA	
Supply Voltage	2.7		5.0	V	
Proof Pressure	2X			Rated	
Burst Pressure	5X			Rated	No More than 20kpsi
Isolation, Body to Any Lead	50			MΩ	@ 250V _{DC}
Pressure Cycles	1.00E+6			0~F.S. Cycles	
Pressure Accuracy (RSS combined Non-Linearity, Hysteresis & Repeatability)	-0.25		0.25	%F.S. BFSL	@ 25°C
Temperature Accuracy	-3		3	°C	Note 2
Long Term Stability (1 year)	-0.25		0.25	%F.S.	
Total Error Band	-1.5		1.5	%F.S.	Over comp Temp.
Compensated Temperature	0		55	°C	
Compensated Temperature Output	512		1075	Count	For reference
Operating Temperature	-20		+85	°C	M12 connector option, 125°C Max
Storage Temperature	-40		+85	°C	M12 connector option, 125°C Max
Response time			3	ms @ 4MHz	Non-sleep mode, note 3
Response time			8.4	ms @ 4MHz	Sleep mode, note 3
Wetted Material (except elastomer seal)	17-4PH Stainless Steel				
Shock	50g, 11 msec Half Sine Shock per MIL-STD-202G, Method 213B, Condition A				
Weather proof Rating ³	IP67				
Vibration	±20g, MIL-STD-810C, Procedure 514.2-2, Curve L				

Compliance⁶

EN 55011 Emissions Class A & B

IEC 61000-4-2 Electrostatic Discharge Immunity (4kV contact/8kV air discharge)

IEC 61000-4-3 Radiated Radio-Frequency Electromagnetic Field Immunity (1V/m, 80M-1GHz; 3 V/m, 1.4 – 2.0GHz; 1V/m, 2.0-2.7GHz)

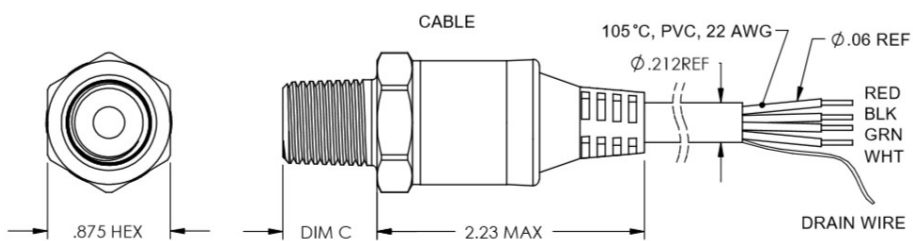
IEC 61000-4-4 Electrical Fast Transient/Burst Immunity (±1kV)

IEC 61000-4-6 immunity to conducted disturbances, induced by radio-frequency fields (150k-80MHz, 3V_{RMS})

Notes

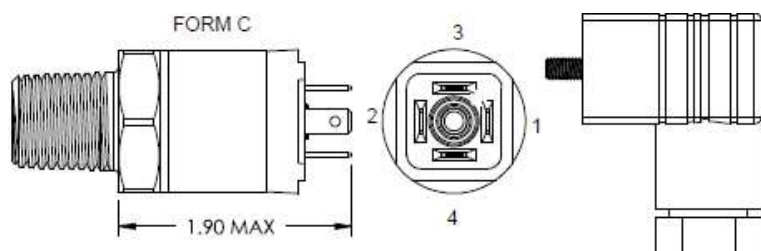
1. Weather-proof ratings are met when the mating connectors are properly installed and cable termination to dry and clean area. For Cable option, IP67 is guaranteed under room temperature.
2. Reflect pressure port diaphragm temperature over the compensated temperature range.
3. Response time is from power on to reading measurement data.
4. For all CE compliance test, max allowed output deviation is ±1.5%F.S.
5. All Configurations are built with Voltage Reverse and output Short-Circuit Protections.
6. For communication and interfacing, refer to document 'Interfacing to MEAS Digital Pressure Modules' online

DIMENSIONS

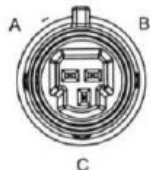
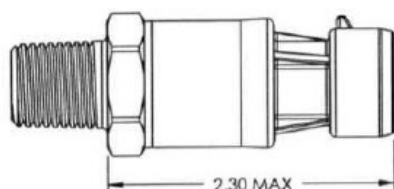


Digital Output I²C Wiring

Connection	+Supply	-Supply	SDA	SCL
Cable	RED	BLACK	GREEN	WHITE
M12	1	3	4	2

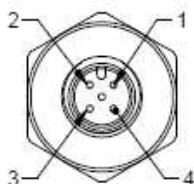
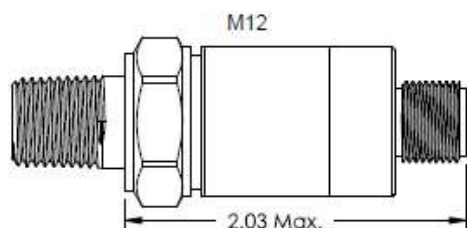


PACKARD CONNECTOR



Current Output Wiring

Connection	+Supply	-Supply	NC. Pins	P _{REF} Vent
Packard A	A	B	C	Hole through connector
Packard B	B	A	C	Hole through connector
Form C	1	2	3, 4	Thread through connector
Cable	Red	Black	-	In Cable
M12	1	3	2, 4	Hole through connector



Voltage Output Wiring

Connection	+Supply	-Supply	+Output	-Output	NC. Pins*	P _{REF} Vent
Packard A	A	B	C	-	-	Hole through Connector
Packard B	B	A	C	-	-	Hole through Connector
Form C	1	2	3	-	4	Thread through Connector
Cable	Red	Black	White	Not connected	-	In Cable
M12	1	3	2		4	Hole through Connector

Notes:

*NC. Pins are reserved for factory use only. **DO NOT CONNECT.**

**For cable connections, drain wire is internally terminated to pressure port.
drain wire is not available for I2C output option

*** Cable material : 4C*22AWG + DRAIN + AL.MYLAR + PVC Jacket

Transmitter of gage pressure type requires vent to atmosphere on the pressure reference side.

- Accomplished via cable from transmitter or through customer mating connector/cable assembly which has internal vent path (end of cable should be terminated to clean & dry area)

Weather-proof Ratings are met when Mating Connectors are installed properly, and cable termination is to dry and clean area.

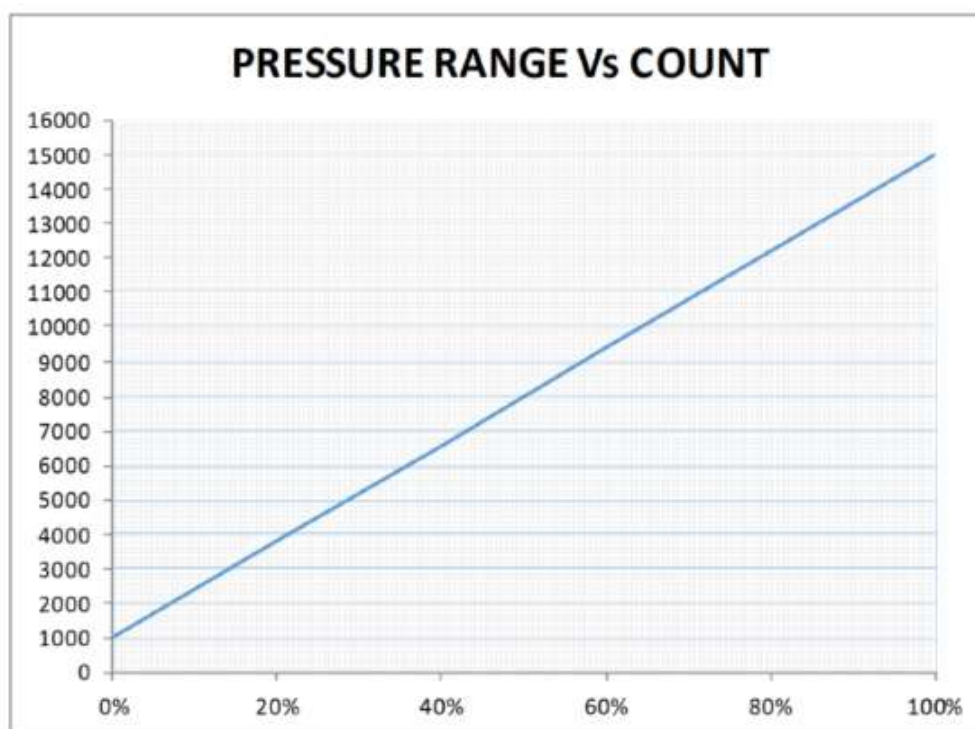
PRESSURE PORTS

Code	Pressure Port	Dim C	Recommended Torque [Nm]
2	1/4-19 BSPP	0.47[11.94]	30-35
4	7/16-20 UNF Male SAE J1926-2 Straight Thread O-Ring BUNA-N 90SH ID8.92xW1.83mm	0.45 [11.43]	18-20
5	1/4-18 NPT	0.65 [16.51]	2-3 TFFT*
6	1/8-27 NPT	0.53 [13.46]	2-3 TFFT*
B	G1/4 JIS B2351 with NBR O-ring	0.47 [11.94]	30-35
E	1/4-19 BSPT	0.50 [12.70]	2-3 TFFT*
P	7/16-20 UNF Female SAE J513 Straight Thread w/ Integral Valve Depressor	0.43 [10.92]	15-16

*Turn From Finger Tight

DIGITAL PRESSURE OUTPUT

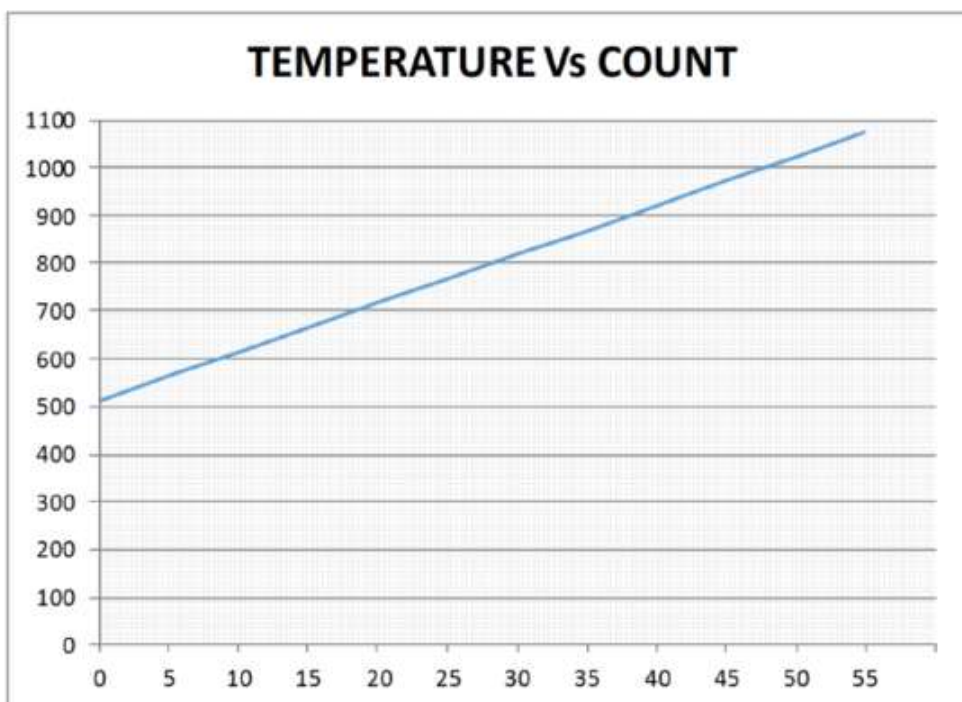
% Output	Digital Counts (Decimal)	Digital Counts (Hex)
0%	1000	0x3E8
5%	1700	0X6A4
10%	2400	0X960
50%	8000	0X1F40
90%	13600	0X3520
95%	14300	0X37DC
100%	15000	0X3A98



$$\text{OUTPUT (DECIMAL COUNTS)} = \frac{15000-1000}{P_{\text{max}}-P_{\text{min}}} \times (P_{\text{applied}}-P_{\text{min}}) + 1000$$

DIGITAL TEMPERATURE OUTPUT

Output °C	Digital Counts (Decimal)	Digital counts
0	512	0x200
10	614	0x266
25	767	0x2FF
40	921	0x399
55	1075	0x433



$$\text{OUTPUT (DECIMAL COUNTS)} = \frac{(\text{OUTPUT } ^\circ\text{C} + 50 ^\circ\text{C}) \times 2048}{150 ^\circ\text{C} - (-50 ^\circ\text{C})}$$

OUTPUT (ANALOG)

Code	Output	Supply	Ratiometricity	Red	Black	Green	White
3	0.5 – 4.5V	5 ± 0.25V	Yes	+Supply	Common	Not connected	+Output
5	4 – 20mA	9 – 30V	No	+Supply	-Supply	Not connected	Not connected
6	0 – 5 V	8 – 30V	No	+Supply	-Supply	Not connected	+Output
7	0 – 10 V	12 – 30 V	No	+Supply	-Supply	Not connected	+Output
8	1 – 5 V	8 – 30 V	No	+Supply	-Supply	Not connected	+Output

OUTPUT (DIGITAL)

Code	Output	Supply	Red	Black	Green	White
J	I ² C	2.7 – 5.0V	+Supply	-Supply	SDA	SCL

ORDERING INFORMATION

For Analog Output:

M32 3 4 - 00000 4 - 250P G

Output	
Code	Output
3	0.5-4.5V
5	4-20mA
6	0-5V
7	0-10V
8	1~5V

Connection	
4	Packard A Connector
6	Form C with Mating Connector
9	Packard B Connector
D	M12 Connector
L	Cable 0.5m
M	Cable 1m
N	Cable 2m
P	Cable 5m

Snubber	
0	No snubber
1	With snubber*

*Available for G1/4 port only, more snubber option, please consult with factory

For Digital Output, see "For Digital Output" Ordering Information
All Configurations are built with Voltage Reverse and Output Short-Circuit Protections.

Pressure Type	
G	Gage
C	Compound

Compound pressure range is -14.7 to XXX psiG or -1 to XXX barG.
i.e. 200PC: -14.7 to 200psiG, 020BC: -1 to 20 barG

Pressure Range	
psi STD	bar STD
100P	007B
250P	017B
500P	035B
01KP	070B
2K5P	170B
05KP	350B
7K5P	500B
10KP	700B

Pressure Ranges between 100-10000psi (7-700bar) are all available. Change Pressure Number Accordingly

Pressure Port	
Code	Description
2	1/4-19 BSPP
4	7/16-20 UNF Male SAE J1926-2 Straight Thread O-ring 90SH ID8.92xW1.83mm
5	1/4-18 NPT
6	1/8-27 NPT
B	G1/4 JIS B2351 with NBR O-ring
E	1/4-19 BSPT
P	7/16-20 UNF Female SAE J513 Straight Thread with Integral Valve Depressor

Click [here](#) for Torque Recommendation

For Digital Output:

M32 J L – 000 0 0 4 – 250P G

Output	
Code	Output
J	I ² C

Connection	
L	Cable 0.5m
M	Cable 1m
D	M12 connector

Snubber	
0	No snubber
1	With snubber*

*Available for G1/4 port only, More snubber options, please consult with factory.

Sleep Mode (Digital ONLY)	
0	Non-Sleep Mode
1	Sleep Mode

Digital Address (Digital ONLY)	
0	0X28H
1	0X36H
2	0X46H
3	0X48H
4	0X51H

Pressure Type	
G	Gage
C	Compound

Compound pressure range is -14.7 to XXX psiG or -1 to XXX barG.
Ex. 200PC: -14.7 to 200psiG, 020BC: -1 to 20 barG

Pressure Range	
psi STD	bar STD
100P	007B
250P	017B
500P	035B
01KP	070B
2K5P	170B
05KP	350B
7K5P	500B
10KP	700B

Pressure Ranges between 100-10000psi (7-700bar) are all available. Change Pressure Number Accordingly

Pressure Port	
Code	Description
2	1/4-19 BSPP
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5	1/4-18 NPT
6	1/8-27 NPT
B	G1/4 JIS B2351 with NBR O-ring
E	1/4-19 BSPT
P	7/16-20 UNF Female SAE J513 Straight Thread with Integral Valve Depressor

Click [here](#) for Torque Recommendation

All Configurations are built with Voltage Reverse and Output Short-Circuit Protections.

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INTERFACING TO TE DIGITAL PRESSURE MODULES

The TE series of digital pressure sensors uses the latest CMOS sensor conditioning circuitry (SSC) to create a low cost, high performance digital output pressure (14-bit) and temperature (11-bit) sensor designed to meet the strictest requirements from OEM customers.

The MS45x5DO, 85BSD, 85FBSD, 86BSD, 154BSD, MSP100(DO) and MSP300(DO), M3200(DO), FX29(DO) and FS30(DO) are the latest offering from TE to offer digital communication to pressure sensor OEMs.

I²C AND SPI INTERFACE SPECIFICATIONS

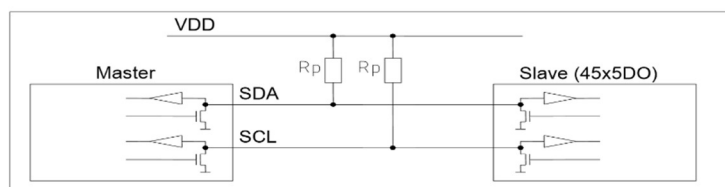
1. I²C Interface Specification

The I²C interface is a simple 8-bit protocol using a serial data line (SDA) and a serial clock line (SCL) where each device connected to the bus is software addressable by a unique address. For detailed specifications of the I²C protocol, see The I²C Bus Specification, Version 2.1, January 2000.

1.1 Interface Connection-External

Bi-directional bus lines are implemented by the devices (master and slave) using open-drain output stages and a pull-up resistor connected to the positive supply voltage. The recommended pull-up resistor value depends on the system setup (capacitance of the circuit or cable and bus clock frequency). In most cases, 4.7k Ω is a reasonable choice. The capacitive loads on SDA and SCL line have to be the same. It is important to avoid asymmetric capacitive loads.

I²C Transmission Start Condition



Both bus lines, SDA and SCL, are bi-directional and therefore require an external pull-up resistor.

1.2 I²C Address

The I²C address consists of a 7-digit binary value. The factory setting for the I²C slave address is 0x28, 0x36 or 0x46 depending on the interface type selected from the ordering information. The address is always followed by a write bit (0) or read bit (1). The default hexadecimal I²C header for read access to the sensor is therefore 0x51, 0x6D, 0x8D respectively, based on the ordering information.

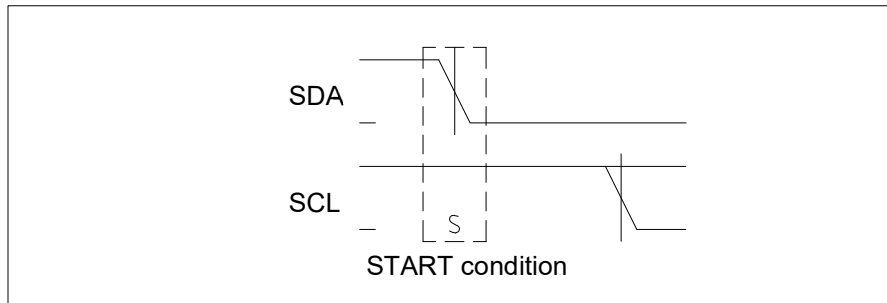
1.3 INT/SS Pin

When programmed as an I²C device, the INT/SS pin operates as an interrupt. The INT/SS pin rises when new output data is ready and falls when the next I²C communication occurs.

1.4 Transfer Sequences

Transmission START Condition (S): The START condition is a unique situation on the bus created by the master, indicating to the slaves the beginning of a transmission sequence (the bus is considered busy after a START).

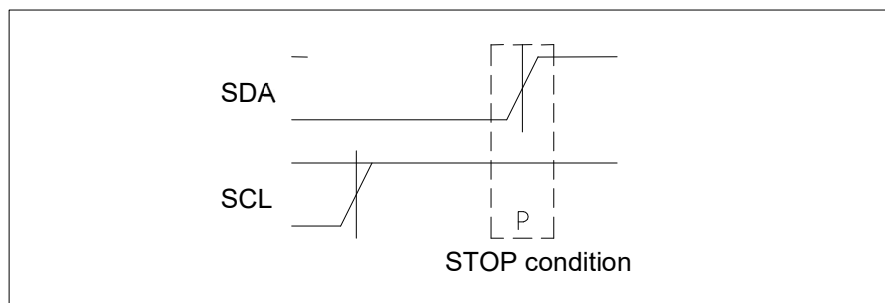
I²C Transmission Start Condition



A HIGH to LOW transition on the SDA line while SCL is HIGH

Transmission STOP Condition (P): The STOP condition is a unique situation on the bus created by the master, indicating to the slaves the end of a transmission sequence (the bus is considered free after a STOP).

I²C Transmission Stop Condition

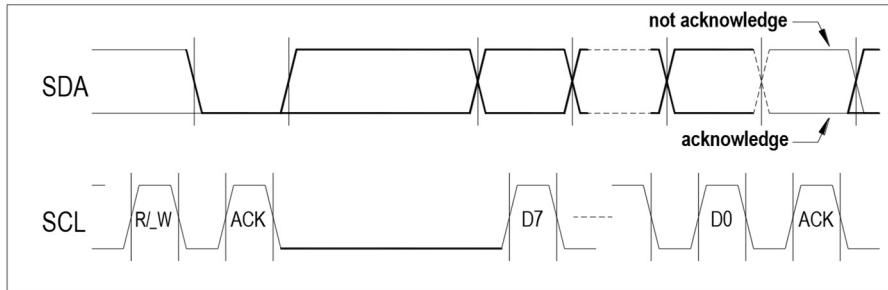


A LOW to HIGH transition on the SDA line while SCL is HIGH

Acknowledge (ACK) / Not Acknowledge (NACK): Each byte (8 bits) transmitted over the I²C bus is followed by an acknowledge condition from the receiver. This means that after the master pulls SCL low to complete the transmission of the 8th bit, SDA will be pulled low by the receiver during the 9th bit time. If after transmission of the 8th bit the receiver does not pull the SDA line low, this is considered to be a NACK condition.

If an ACK is missing during a slave to master transmission, the slave aborts the transmission and goes into idle mode.

I²C ACKNOWLEDGE / NOT ACKNOWLEDGE



Each byte is followed by an acknowledge or a not acknowledge, generated by the receiver

1.5 Data Transfer Format

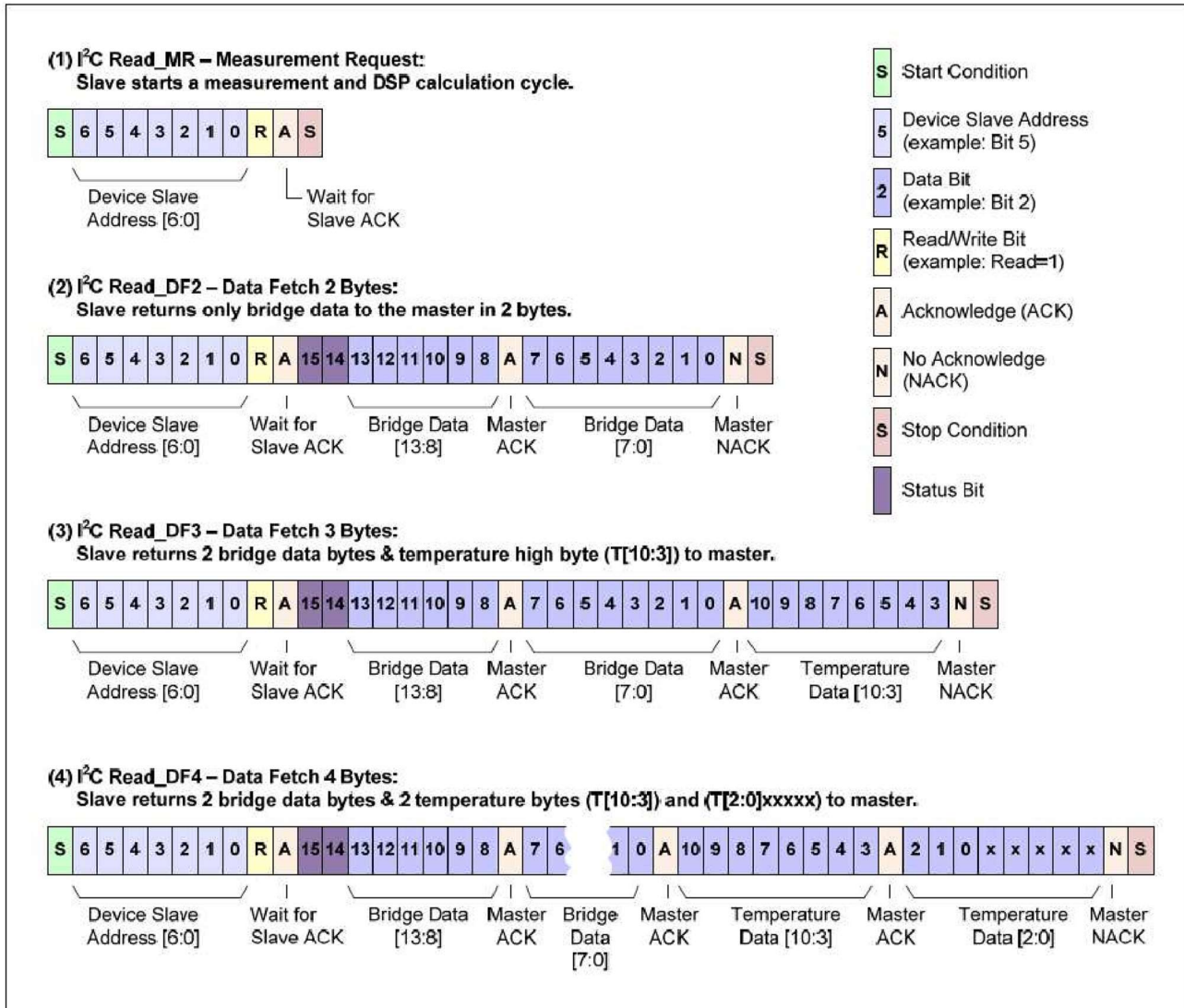
Data is transferred in byte packets in the I²C protocol, which means in 8-bit frames. Each byte is followed by an acknowledge bit. Data is transferred with the most significant bit (MSB) first.

A data transfer sequence is initiated by the master generating the Start condition (S) and sending a header byte. The I²C header consists of the 7-bit I²C device address and the data direction bit (R/_W).

The value of the R/_W bit in the header determines the data direction for the rest of the data transfer sequence. If R/_W = 0 (WRITE), the direction remains master-to-slave, while if R/_W = 1 (READ), the direction changes to slave-to-master after the header byte.

1.6 Command Set and Data Transfer Sequences

The I²C master command starts with the 7-bit slave address with the 8th bit = 1 (READ). The sensor acts as the slave and sends an acknowledge (ACK) indicating success. The sensor has four I²C read commands: Read_MR, Read_DF2, Read_DF3, and Read_DF4. Figure 1.6 shows the structure of the measurement packet of the four I²C read commands, which are explained in sections 1.6.1.



1.6.1 Figure 1.6 – I²C Measurement Packet Reads I²C Read_DF (Data Fetch)

For Data Fetch commands, the number of data bytes returned by the sensor, is determined when the master sends the NACK and stop condition. For the Read_DF3 data fetch command (Data Fetch 3 Bytes; see example 3 in Figure 1.6), the sensor returns three bytes in response to the master sending the slave address and the READ bit (1): two bytes of bridge data with the two status bits as the MSBs and then 1 byte of temperature data (8-bit accuracy). After receiving the required number of data bytes, the master sends the NACK and stop condition to terminate the read operation. For the Read_DF4 command, the master delays sending the NACK and continues reading an additional final byte to acquire the full corrected 11-bit temperature measurement. In this case, the last 5 bits of the final byte of the packet are undetermined and should be masked off in the application. The Read_DF2 command is used if corrected temperature is not required. The master terminates the READ operation after the two bytes of bridge data (see example 2 in Figure 1.6).

The two status bits (Bit 15 and Bit 14) give an indication of stale or valid data depending on their value. A returned value of 00 indicate "normal operation and a good data packet" while a returned value of 10 indicates "stale data that has been already fetched". See section 1.7 for additional details. Users that use "status bit" polling should select a frequency slower than 20% more than the update time.

1.7 Status Bits and Diagnostic Features

The table below summarizes the status bits conditions indicated by the 2 MSBs (Bit (15:14) of I²C data packet, S(1:0) of SPI data packet of the bridge high byte data.

Table 1: Status Bits Encoding

Status Bits (2 MSB of Output Data Packet)	Definition
00	Normal Operation. Good Data Packet
01	Reserved
10	Stale Data. Data has been fetched since last measurement cycle.
11	Fault Detected

The SSC is has on board diagnostic features to ensure robust system operation in the most “mission-critical” applications. A status bit value of “11” indicates a fault condition in the SSC or sensing element. All diagnostics are detected in the next measurement cycle and reported in the subsequent data fetch. Once a diagnostic is reported, the diagnostic status bits will not change unless both the cause of the diagnostic is fixed and a power-on-reset is performed.

1.8 I²C Protocol Differences

There are three differences in the described above protocol compared with original I²C protocol:

- Sending a start-stop condition without any transitions on the SCL line (no clock pulses in between) creates a communication error for the next communication, even if the next start condition is correct and the clock pulse is applied. An additional start condition must be sent, which results in restoration of proper communication.
- The restart condition – a falling SDA edge during data transmission when the SCL clock line is still high – creates the same situation. The next communication fails, and an additional start condition must be sent for correct communication.
- A falling SDA edge is not allowed between the start condition and the first rising SCL edge. If using an I²C address with the first bit 0, SDA must be held down from the start condition through the first bit.

2. SPI Interface Specification

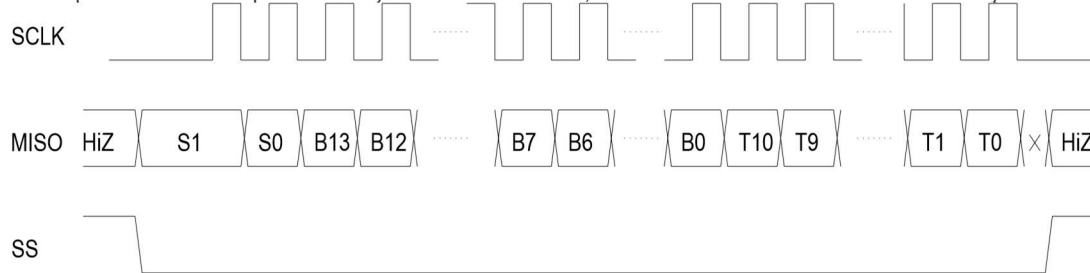
SPI is a general-purpose synchronous serial interface. During an SPI transfer, transmit and receive data is simultaneously shifted out and in serially. A serial clock line synchronizes the shifting and sampling of the information on two serial data lines.

SPI devices communicate using a master-slave relationship. Due to its lack of built-in device addressing, SPI requires more effort and more hardware resources than I²C when more than one slave is involved. But SPI tends to be simpler and more efficient than I²C in point-to-point (single master, single slave) applications for the very same reason; the lack of device addressing means less overhead.

The SPI interface is programmed for falling-edge MISO change.

2.1 SPI Read_DF (Data Fetch)

The SPI interface will have data change after the falling edge of SCLK. The master should sample MISO on the rise of SCLK. The entire output packet is 4 bytes (32 bits). The high bridge data byte comes first, followed by the low bridge data byte. Then 11 bits of corrected temperature (T[10:0]) are sent: first the T[10:3] byte and then the {T[2:0],xxxx} byte. The last 5 bits of the final byte are undetermined and should be masked off in the application. If the user only requires the corrected bridge value, the read can be terminated after the 2nd byte. If the corrected temperature is also required but only at an 8-bit resolution, the read can be terminated after the 3rd byte is read.



Packet = [{S(1:0),B(13:8)}, {B(7:0)}, {T(10:3)}, {T(2:0),xxxx}] Where

S(1:0) = Status bits of packet (normal, command, busy, diagnostic) B(13:8) = Upper 6 bits of 14-bit bridge data.

B(7:0) = Lower 8 bits of 14-bit bridge data.

T(10:3) = Corrected temperature data (if application does not require corrected temperature, terminate read early) T(2:0),xxxx =

Remaining bits of corrected temperature data for full 11-bit resolution

HiZ = High impedance

Figure 2.2 – SPI Output Packet with Falling Edge SPI_Polarity

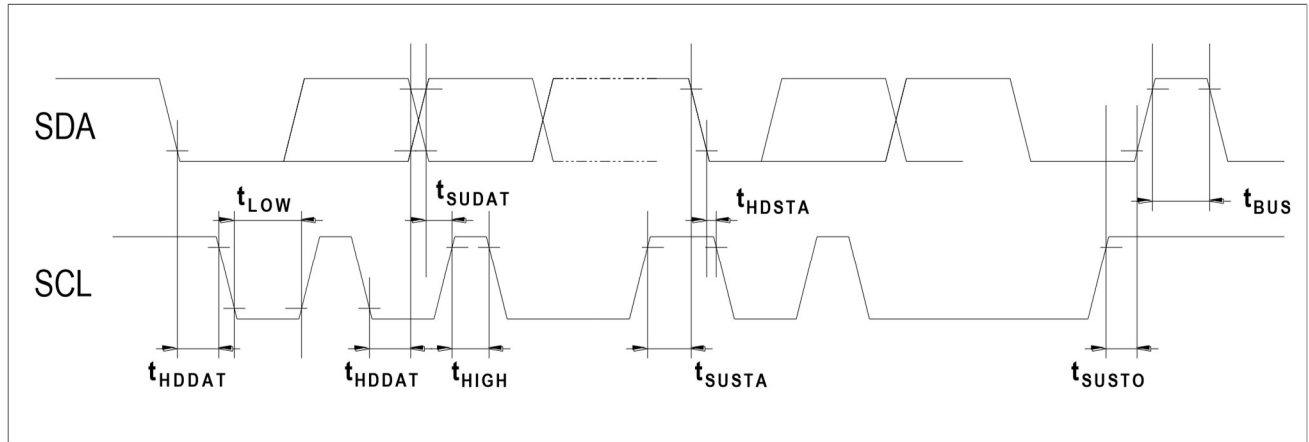
TIMING DIAGRAMS

I²C INTERFACE PARAMETERS

PARAMETERS	SYMBOL	MIN	TYP	MAX	UNITS
SCLK CLOCK FREQUENCY	fsCL	100		400	KHz
START CONDITION HOLD TIME RELATIVE TO SCL EDGE	tHSTA	0.1			uS
MINIMUM SCL CLOCK LOW WIDTH ¹	tLOW	0.6			uS
MINIMUM SCL CLOCK HIGH WIDTH ¹	tHIGH	0.6			uS
START CONDITION SETUP TIME RELATIVE TO SCL EDGE	tsUSTA	0.1			uS
DATA HOLD TIME ON SDA RELATIVE TO SCL EDGE	tHDDAT	0			uS
DATA SETUP TIME ON SDA RELATIVE TO SCL EDGE	tsUDAT	0.1			uS
STOP CONDITION SETUP TIME ON SCL	tsUSTO	0.1			uS
BUS FREE TIME BETWEEN STOP AND START CONDITION	tBUS	2			uS

¹COMBINED LOW AND HIGH WIDTHS MUST EQUAL OR EXCEED MINIMUM SCL PERIOD.

I2C Timing Diagram

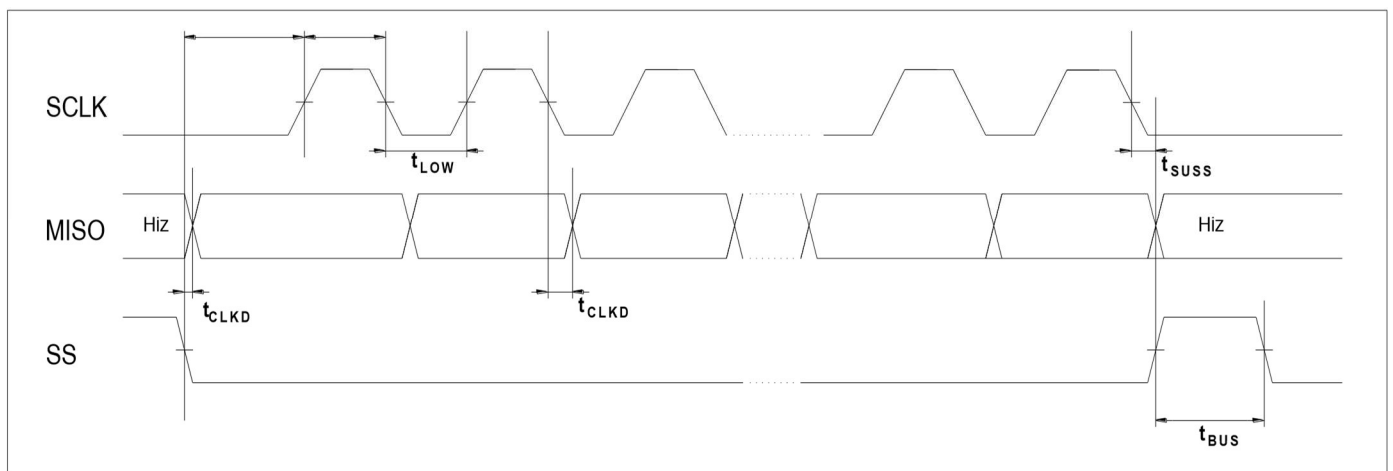


SPI INTERFACE PARAMETERS

PARAMETERS	SYMBOL	MIN	TYP	MAX	UNITS
SCLK CLOCK FREQUENCY	f_{SCL}	50		800	KHz
SS DROP TO FIRST CLOCK EDGE	t_{HDSS}	2.5			μ S
MINIMUM SCL CLOCK LOW WIDTH ¹	t_{LOW}	0.6			μ S
MINIMUM SCL CLOCK HIGH WIDTH ¹	t_{HIGH}	0.6			μ S
CLOCK EDGE TO DATA TRANSITION	t_{CLKD}	0		0.1	μ S
RISE OF SS RELATIVE TO LAST CLOCK EDGE	t_{SUSS}	0.1			μ S
BUS FREE TIME BETWEEN RISE AND FALL OF SS	t_{BUS}	2			μ S

¹ COMBINED LOW AND HIGH WIDTHS MUST EQUAL OR EXCEED MINIMUM SCLK PERIOD.

SPI TIMING DIAGRAM



C Code Example For FX29

//Note: The C code is use for communication with FX29K0-040B-0100-L using STM32L031.

// This routine is applicable to other models mentioned in this document.

```
#include "main.h"
```

```
#include "stm32l0xx_hal.h"
```

```
#include "stdlib.h"
```

```
#include "delay.h"
```

```
#include "config.h"
```

```
u8 temp[7];
```

```
float Tscope,Pscope,Tdisplay,Pdisplay;
```

```
float Lmax=100,Lmin=0 ; //Span 100L, Zero 0L, Span should be defined by the sensor  
pressure range of customer used. 100 means pressure range of 100L
```

```
u32 Pvalue,Tvalue,Tspan,Pspan;
```

```
u16 P1=1000,P2=15000;
```

```
void SDA_IN2(void);
```

```
void SDA_OUT2(void);
```

```
void IIC_Start2(void);
```

```
void IIC_Stop2(void);
```

```
unsigned char IIC_Wait_Ack2(void);
```

```
void IIC_Ack2(void);
```

```
void IIC_NAck2(void);
```

```
void IIC_Send_Byte(unsigned char txd);
```

```
unsigned char IIC_Read_Byte(unsigned char ack);
```

```
float Get_I2CValue(void);
```

```
void SDA_IN2()
```

```
{
```

```
    GPIO_InitTypeDef GPIO_InitStructure;
```

```
    GPIO_InitStructure.Pin = SDA2_Pin;
```

```
    GPIO_InitStructure.Mode = GPIO_MODE_INPUT;
```

```
    GPIO_InitStructure.Pull = GPIO_NOPULL;
```

```
    //GPIO_InitStructure.Alternate = GPIO_PuPd_UP;
```

```
    GPIO_InitStructure.Speed = GPIO_SPEED_FREQ_LOW;
```

```
    HAL_GPIO_Init(SDA2_GPIO_Port, &GPIO_InitStructure);
```

```
}
```

```
void SDA_OUT2()
```

```
{
```

```
    GPIO_InitTypeDef GPIO_InitStructure;
```

```

    GPIO_InitStructure.Pin = SDA2_Pin;
    GPIO_InitStructure.Mode = GPIO_MODE_OUTPUT_PP;
    GPIO_InitStructure.Pull = GPIO_NOPULL;
    GPIO_InitStructure.Speed = GPIO_SPEED_FREQ_LOW;
    HAL_GPIO_Init(SDA2_GPIO_Port, &GPIO_InitStructure);
}
void IIC_Start2()
{
    SDA_OUT2(); //sda???
    Sensor_SDA_ON ;
    Sensor_SCL_ON;
    delay_us(4);
    Sensor_SDA_OFF;//START:when CLK is high,DATA change form high to low
    delay_us(4);
    Sensor_SCL_OFF;//??I2C??,?????????
}
void IIC_Stop2()
{
    SDA_OUT2();//sda???
    Sensor_SCL_OFF;
    Sensor_SDA_OFF;//STOP:when CLK is high DATA change form low to high
    delay_us(4);
    Sensor_SCL_ON;
    Sensor_SDA_ON ;//??I2C??????
    delay_us(4);
}
unsigned char IIC_Wait_Ack2()
{
    unsigned char ucErrTime=0;
    SDA_IN2(); //SDA?????
    Sensor_SDA_ON ;delay_us(1);
    Sensor_SCL_ON;delay_us(1);
    while(READ_Sensor_SDA)
    {
        ucErrTime++;
        if(ucErrTime>250)
        {
            IIC_Stop2();
            return 1;
        }
    }
}

```

```
    }
    Sensor_SCL_OFF;//????0
    return 0;
}
void IIC_Ack2()
{
    Sensor_SCL_OFF;
    SDA_OUT2();
    Sensor_SDA_OFF;
    delay_us(2);
    Sensor_SCL_ON;
    delay_us(2);
    Sensor_SCL_OFF;
}
void IIC_NAck2()
{
    Sensor_SCL_OFF;
    SDA_OUT2();
    Sensor_SDA_ON;
    delay_us(2);
    Sensor_SCL_ON;
    delay_us(2);
    Sensor_SCL_OFF;
}
void IIC_Send_Byte(unsigned char txd)
{
    unsigned char t;
        SDA_OUT2();
    Sensor_SCL_OFF;//????????????
    for(t=0;t<8;t++)
    {
        if(txd&0x80)
            {Sensor_SDA_ON;}
        else
            {Sensor_SDA_OFF;}

        txd<<=1;
        delay_us(2); //?TEA5767????????????
        Sensor_SCL_ON;
        delay_us(2);
        Sensor_SCL_OFF;
```

```

        delay_us(2);
    }
}
unsigned char IIC_Read_Byte(unsigned char ack)
{
    unsigned char i, receive=0;
    SDA_IN2();//SDA?????
    for(i=0; i<8; i++)
    {
        Sensor_SCL_OFF;
        delay_us(2);
        Sensor_SCL_ON;
        receive<<=1;
        if(READ_Sensor_SDA) receive++;
        delay_us(1);
    }
    if (!ack)
        IIC_NAck2();//??nACK
    else
        IIC_Ack2();//??ACK
    return receive;
}
u8 I2C_ERR=0;
float Get_I2CValue()
{
    //Wake_up, if non-sleep mode this part is no needed.
    IIC_Start2();          //MR command
    IIC_Send_Byte(0x51);
    IIC_Wait_Ack2();
    IIC_Stop2();
    HAL_Delay(2);          //2ms
    delay

    //////////////////////////////////////

    IIC_Start2();          //DF4
    IIC_Send_Byte(0x51);
    IIC_Wait_Ack2();
    temp[0]=IIC_Read_Byte(1);
    temp[1]=IIC_Read_Byte(1);
    temp[2]=IIC_Read_Byte(1);

```

```
temp[3]=IIC_Read_Byte(0);
IIC_Stop2();
if((temp[0]&0xc0)==0x00)
{
    Pvalue=(temp[0]<<8) | temp[1];
    Tvalue=(temp[2]<<3) | (temp[3]>>5);
    I2C_ERR=0;
}
else
    I2C_ERR=1;

Tscope=200;//-50~150
Tspan=2048;//11bit

if(I2C_ERR==0)
{
    Pspan=P2-P1;
    Tdisplay=Tvalue*Tscope/Tspan-50;
    Pdisplay=Pvalue*(Lmax-Lmin)/Pspan+Lmin;//100L
}
return Pdisplay;
```

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